



Install & Import Libraries

```
In [1]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [23]: import torch
import torch.nn as nn
import torch.optim as optim
import torchvision.transforms as transforms
import torchvision.models as models
from PIL import Image
import matplotlib.pyplot as plt
import os
```

Device Configuration

```
In [24]: device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)
```

Using device: cuda

```
In [25]: BASE_DIR = "/content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/"
```

Image Loader

```
In [27]: image_size = 512 if torch.cuda.is_available() else 256

loader = transforms.Compose([
    transforms.Resize((image_size, image_size)),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std=[0.229, 0.224, 0.225]
    )
])

def load_image(image_path):
    image = Image.open(image_path).convert("RGB")
    image = loader(image).unsqueeze(0)
    return image.to(device, torch.float)

content_image = load_image(BASE_DIR+"eye.jpg")
style_image = load_image(BASE_DIR+"style2.jpg")
```

Display Images

```
In [37]: def imshow(tensor, title=None):
          image = tensor.clone().detach().cpu().squeeze(0)

          # De-normalize
          mean = torch.tensor([0.485, 0.456, 0.406]).view(3,1,1)
          std = torch.tensor([0.229, 0.224, 0.225]).view(3,1,1)

          image = image * std + mean
          image = torch.clamp(image, 0, 1)

          image = transforms.ToPILImage()(image)

          plt.imshow(image)
          if title:
              plt.title(title)
          plt.axis("off")
```

```
In [38]: plt.figure(figsize=(10,5))

          plt.subplot(1,2,1)
          imshow(content_image, "Content Image")

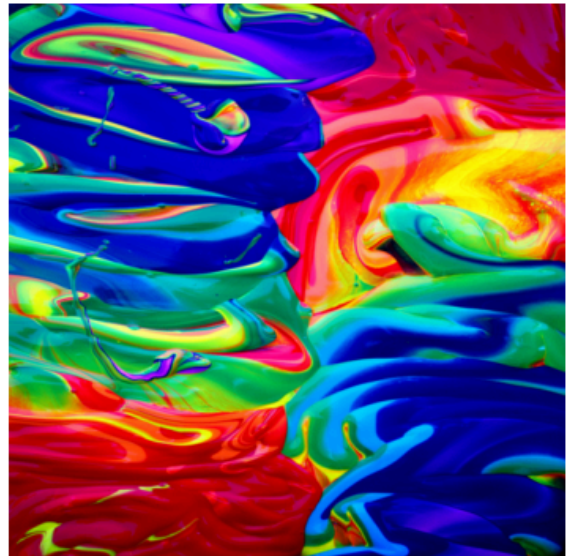
          plt.subplot(1,2,2)
          imshow(style_image, "Style Image")

          plt.show()
```

Content Image



Style Image



Load Pretrained VGG19 Model

```
In [39]: vgg = models.vgg19(pretrained=True).features.to(device).eval()

for param in vgg.parameters():
    param.requires_grad = False
```

Feature Extraction

```
In [40]: def get_features(image, model):
    layers = {
        '0': 'conv_1',
        '5': 'conv_2',
        '10': 'conv_3',
        '19': 'conv_4',
        '28': 'conv_5'
    }

    features = {}
    x = image

    for name, layer in model._modules.items():
        x = layer(x)
        if name in layers:
            features[layers[name]] = x

    return features

content_features = get_features(content_image, vgg)
style_features = get_features(style_image, vgg)
```

Gram Matrix

```
In [41]: def gram_matrix(tensor):
    _, d, h, w = tensor.size()
    tensor = tensor.view(d, h*w)
    gram = torch.mm(tensor, tensor.t())
    return gram
```

Target Image

```
In [42]: target = content_image.clone().requires_grad_(True).to(device)
```

Optimizer & Weights

```
In [43]: optimizer = optim.LBFGS([target])

content_layer = 'conv_4'

style_layers = ['conv_1', 'conv_2', 'conv_3', 'conv_4', 'conv_5']

style_weights = {
    'conv_1': 1.0,
    'conv_2': 0.8,
    'conv_3': 0.5,
    'conv_4': 0.3,
    'conv_5': 0.1
}

content_weight = 5e3
style_weight = 2e5
```

```
In [44]: output_dir = BASE_DIR+"nst_outputs"
os.makedirs(output_dir, exist_ok=True)

print("Folder created:", output_dir)
```

Folder created: /content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs

Training Loop

```
In [45]: steps = 2000
run = [0]

while run[0] <= steps:

    def closure():
        optimizer.zero_grad()

        target_features = get_features(target, vgg)

        content_loss = torch.mean(
            (target_features[content_layer] - content_features[content_layer])
        )

        style_loss = 0
        for layer in style_layers:
            target_feature = target_features[layer]
            target_gram = gram_matrix(target_feature)

            style_feature = style_features[layer]
            style_gram = gram_matrix(style_feature)
```

```

        _, d, h, w = target_feature.shape

        layer_style_loss = style_weights[layer] * torch.mean(
            (target_gram - style_gram)**2
        ) / (d*h*w)

        style_loss += layer_style_loss

    total_loss = content_weight * content_loss + style_weight * style_loss

    total_loss.backward()

    if run[0] % 100 == 0:
        print(f"Step {run[0]}, Total Loss: {total_loss.item()}")
        temp = target.clone().detach().cpu().squeeze(0)
        mean = torch.tensor([0.485, 0.456, 0.406]).view(3,1,1)
        std = torch.tensor([0.229, 0.224, 0.225]).view(3,1,1)
        temp = temp * std + mean
        temp = torch.clamp(temp, 0, 1)

        temp_img = transforms.ToPILImage()(temp)
        temp_img.save(os.path.join(output_dir, f"output_step_{run[0]}.jpg"))

    run[0] += 1
    return total_loss

optimizer.step(closure)

with torch.no_grad():
    target.clamp_(-3, 3)

```

```

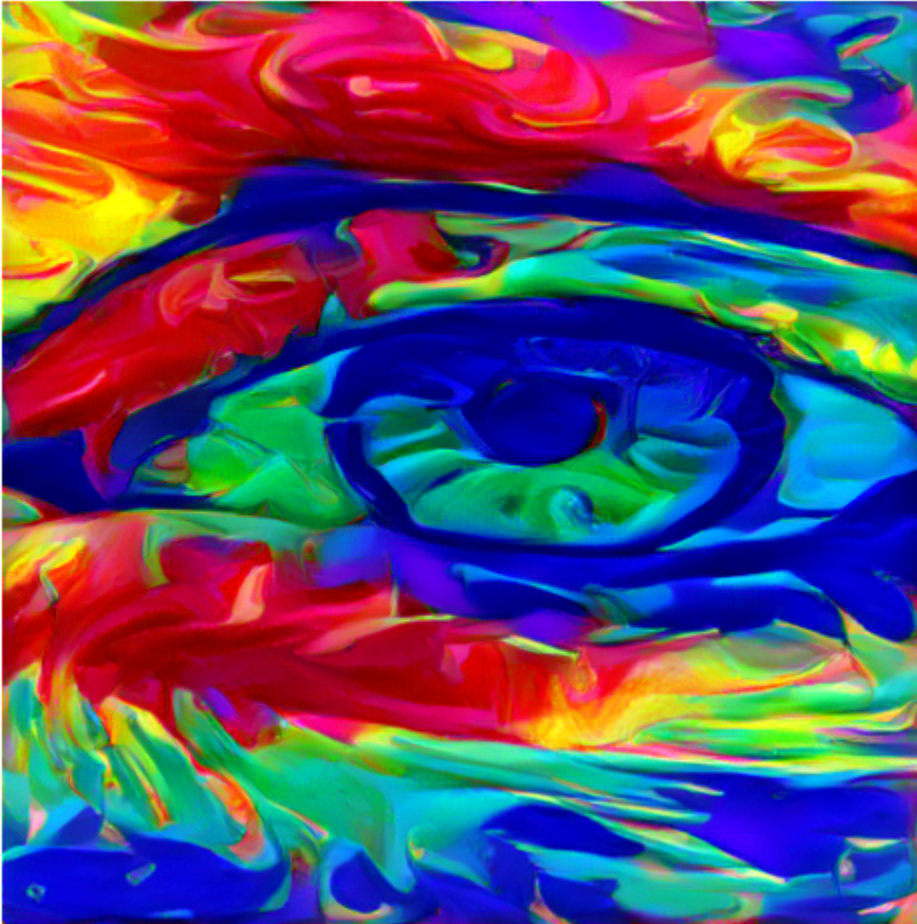
Step 0, Total Loss: 1946905216.0
Step 100, Total Loss: 27933610.0
Step 200, Total Loss: 6035279.0
Step 300, Total Loss: 2708783.5
Step 400, Total Loss: 1049756.375
Step 500, Total Loss: 625699.5625
Step 600, Total Loss: 442300.4375
Step 700, Total Loss: 363842.0625
Step 800, Total Loss: 318630.125
Step 900, Total Loss: 282305.25
Step 1000, Total Loss: 262043.5625
Step 1100, Total Loss: 247757.390625
Step 1200, Total Loss: 238081.28125
Step 1300, Total Loss: 230228.96875
Step 1400, Total Loss: 224083.84375
Step 1500, Total Loss: 219235.015625
Step 1600, Total Loss: 215160.25
Step 1700, Total Loss: 211862.0
Step 1800, Total Loss: 209017.890625
Step 1900, Total Loss: 206767.9375
Step 2000, Total Loss: 204734.921875

```

Show Final Result

```
In [46]: plt.figure(figsize=(6,6))  
         imshow(target, "Stylized Output")  
         plt.show()
```

Stylized Output



Save Final Output

```
In [47]: final = target.clone().detach().cpu().squeeze(0)  
  
         mean = torch.tensor([0.485, 0.456, 0.406]).view(3,1,1)  
         std = torch.tensor([0.229, 0.224, 0.225]).view(3,1,1)  
  
         final = final * std + mean  
         final = torch.clamp(final, 0, 1)  
  
         final_img = transforms.ToPILImage()(final)  
         final_img.save(os.path.join(output_dir, "final_output.jpg"))
```

```
print("Final image saved.")
```

Final image saved.

```
In [48]: !zip -r nst_outputs.zip "/content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/ (store d 0%)
```

```
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/ (store d 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_0.jpg (deflated 1%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_100.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_200.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_300.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_400.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_500.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_600.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_700.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_800.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_900.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1000.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1100.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1200.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1300.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1400.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1500.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1600.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1700.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1800.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_1900.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/output_step_2000.jpg (deflated 0%)
updating: content/drive/MyDrive/Colab Notebooks/GenAI/Lab 7/nst_outputs/final_output.jpg (deflated 0%)
```

```
In [ ]:
```